

Spinning Pendulum

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1 Spherical Pendulum Equations of Motion

Here, we want to consider the effect of some angular velocity ω on the motion of a spherical pendulum. In order to do so, we introduce a slightly modified set of spherical coordinates, where the ωt term is introduced to represent the motion of the pendulum about a vertical axis:

$$x = l \sin(\varphi) \cos(\theta + \omega t) \quad y = l \sin(\varphi) \sin(\theta + \omega t) \quad z = -l \cos(\varphi)$$

Taking the time derivative, keeping in mind that both φ and θ are functions of time:

$$\dot{x} = l \left(\cos(\theta + \omega t) \cos(\varphi) \dot{\varphi} - (\omega + \dot{\theta}) \sin(\theta + \omega t) \sin(\varphi) \right)$$

$$\dot{y} = l \left(\sin(\theta + \omega t) \cos(\varphi) \dot{\varphi} + (\omega + \dot{\theta}) \cos(\theta + \omega t) \sin(\varphi) \right)$$

$$\dot{z} = -l \dot{\varphi} \sin(\varphi)$$

This enables us to write the kinetic energy as follows:

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = \frac{1}{2} m l^2 \left((\dot{\theta} + \omega)^2 \sin^2(\varphi) + \dot{\varphi}^2 \right)$$

Ultimately allowing us to write the Lagrangian as:

$$\mathcal{L} = T - U = \frac{1}{2} m l^2 \left((\dot{\theta} + \omega)^2 \sin^2(\varphi) + \dot{\varphi}^2 \right) - U$$

This is all well and good, now we can write the potential due to gravity as follows; however, this is not the only force which we must consider when writing the equations of motion for a pendulum spinning about its vertical axis. Indeed,

the source of the vertical elevation which we hope to investigate is the fictitious "centripetal" force.

$$U_g = -mgl \cos(\varphi)$$

Although the centripetal force is not often written in terms of a potential, and rather in terms of the rotation vectors and axes, we have use $\vec{F} = -\vec{\nabla}U$ in order to assign a potential to it.

$$\vec{F}_c = m\omega \times (\omega \times \vec{r})$$

$$\vec{\omega} \times (\vec{\omega} \times \vec{r}) = \vec{\omega} (\vec{\omega} \cdot \vec{r}) - \vec{r} (\vec{\omega} \cdot \vec{\omega})$$

ω and $\vec{r}$ are orthogonal, so:

$$\vec{F}_c = -m\vec{r}\omega^2$$

$$F_c = m\omega^2 r$$

$$\vec{F} = -\nabla U$$

$$U_c = - \int F_c dr = -\frac{1}{2}m\omega^2 r^2 = -\frac{1}{2}m\omega^2 l^2 \sin^2(\varphi)$$

Where here we have used $r \rightarrow -l \sin(\varphi)$ to represent the radial direction of rotation. We can then write an effective potential, which includes the centripetal force.

$$U_{eff} = U_g + U_c = -mgl \cos(\varphi) - \frac{1}{2}m\omega^2 l^2 \sin^2(\varphi)$$

$$\mathcal{L} = \frac{1}{2}ml^2 \left((\dot{\theta} + \omega)^2 \sin^2(\varphi) + \dot{\varphi}^2 \right) + mgl \cos(\varphi) + \frac{1}{2}m\omega^2 l^2 \sin^2(\varphi)$$

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\varphi}} \right) = 0$$

$$\frac{\partial L}{\partial \varphi} = ml^2 \sin(\varphi) \cos(\varphi) \left((\dot{\theta} + \omega)^2 + \omega^2 \right) - mgl \sin(\varphi) \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\varphi}} \right) = ml^2 \ddot{\varphi}$$

$$\ddot{\varphi} = \left((\dot{\theta} + \omega)^2 + \omega^2 \right) \sin(\varphi) \cos(\varphi) - \frac{g}{l} \sin(\varphi)$$

$$\frac{\partial \mathcal{L}}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = 0$$

$$\frac{\partial L}{\partial \theta} = 0 \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{d}{dt} \left(ml^2 (\dot{\theta} + \omega) \sin^2(\varphi) \right) = ml^2 \left(\ddot{\theta} \sin^2(\varphi) + 2 (\dot{\theta} + \omega) \dot{\varphi} \sin(\varphi) \cos(\varphi) \right)$$

$$\ddot{\theta} + 2 (\dot{\theta} + \omega) \dot{\varphi} \cot(\varphi) = 0$$